

Changes in Electroencephalographic Bicoherence During Sevoflurane Anesthesia Combined with Intravenous Fentanyl

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With the introduction of bispectral index (BIS) as a measure of a patient's sedation during general anesthesia, attention has been directed toward bispectral analysis of electroencephalography (EEG). In the present study we evaluated the relationship between EEG bicoherence and sevoflurane concentration. Sixteen ASA physical status I–II patients scheduled for elective abdominal surgery were enrolled in the study. Anesthesia was induced with 5% sevoflurane and maintained with sevoflurane and oxygen (50%). Just before surgery, IV fentanyl ($2 \mu\text{g}/\text{kg}$) was given and then continuously infused ($2 \mu\text{g} \cdot \text{kg}^{-1} \cdot \text{h}^{-1}$). Using software we developed, EEG bicoherence, BIS, and 95% spectral edge frequency (SEF95) were recorded at end-tidal sevoflurane concentrations of 0.5%, 0.8%, 1.1%, 1.4%, 1.7%, 2.0%, and 2.3%. Under light anesthesia, EEG bicoherence values were low. With increasing sevoflurane concentrations, 2 peaks of bicoherence emerged along the diagonal line ($f_1 = f_2$). Both the first (at around 4 Hz) and second (at around 10 Hz) grew higher ($37.7\% \pm 7.5\%$ and $35.1\% \pm 9.0\%$, respectively) as the sevoflurane concentration increased to 1.4%. However, the first peak leveled off whereas the second tended to decrease slightly with further increases in sevoflurane concentration. The BIS value decreased as the sevoflurane concentration increased and leveled off at 1.4% and higher concentrations of sevoflurane. The SEF 95 also decreased as the sevoflurane concentration increased up to 2.3%. Thus the distribution pattern of the two bicoherence peaks is likely to be better than BIS of the anesthetic effect of sevoflurane during surgery.

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With the introduction of the bispectral index (BIS) as a measure of sedation during general anesthesia, attention has been focused on bispectral analysis of electroencephalography (EEG), which is the core technology of BIS monitoring (1). The principle of bispectral analysis was described in detail by Sigl and Chamoun (2). In brief, when two wave components with frequency f_1 and f_2 are given to a neural circuit, an output signal with frequency $f_1 + f_2$ is generated. If phase angles of the original components, ϑ_1 and ϑ_2 are inherent to the generated signal (expressed as $\vartheta_1 + \vartheta_2$), then these components are called “phase coupled.” Bispectral analysis quantifies the degree of phase coupling between the signal components with frequencies of f_1 , f_2 , and $f_1 + f_2$ in the signals.

Bicoherence, the normalized bispectrum, is an indicator of the amount of phase coupling. However, the BIS

uses the bispectrum rather than bicoherence to assess the depth of anesthesia (1). We have developed a software tool, the Bispectrum Analyzer (BSA), to perform real-time bispectral analysis of EEG (3). With the BSA, we investigated the changes in EEG bicoherence during isoflurane anesthesia (4). EEG bicoherence values were low in patients under light isoflurane anesthesia. At an increased concentration of isoflurane, two peaks of EEG bicoherence were observed along the diagonal line ($f_1 = f_2$). The first peak (around 4 Hz) and the second peak (at around 10 Hz) increased with increasing isoflurane concentrations. However, although the first peak reached a plateau, the second decreased slightly at higher isoflurane concentrations. However, we did not measure BIS in that study. Whether EEG bicoherence occurs during anesthesia with other anesthetics is unknown. In the present study, we evaluated the change in EEG bicoherence during sevoflurane anesthesia combined with IV fentanyl and compared it with other EEG parameters such as BIS and spectral edge frequency (SEF).

METHODS

The study protocol was approved by the IRB of our institution and written informed consent was obtained from each patient. Sixteen ASA physical status I–II patients scheduled for elective abdominal surgery were enrolled in the study.

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The patients were premedicated with 0.5 mg of atropine administered IM approximately 30 min before the start of anesthesia. Anesthesia was induced with 5% sevoflurane. After tracheal intubation, anesthesia was maintained with sevoflurane in oxygen (50%) and nitrogen. Vecuronium was given as required. Fentanyl ($2 \mu\text{g}/\text{kg}$) was given IV just before the start of surgery and followed by continuous infusion ($2 \mu\text{g} \cdot \text{kg}^{-1} \cdot \text{h}^{-1}$) until the termination of data collection. Mechanical ventilation was adjusted to maintain an end-tidal carbon dioxide partial pressure of 35–40 mm Hg. Noninvasive mean arterial blood pressure (MAP), heart rate (HR), and pulse oximetry (SpO_2) were monitored continuously and maintained within normal ranges (MAP > 60 mm Hg, $50 < \text{HR} < 100$, $\text{SpO}_2 > 95\%$). To maintain MAP above 60 mm Hg, as required, $100 \mu\text{g}$ of phenylephrine was administered. The bladder temperature was monitored continuously and the patients were maintained at normothermia (36.0°C – 36.5°C) using a water blanket (Medi-Therm II, Gaymer, New York, NY).

Before the induction of anesthesia, the electrode for measuring BIS (Bis Sensor, Aspect Medical Systems, Natick, MA) was applied to the forehead regions. The EEG was monitored continuously using an Aspect A-1050 monitor (BIS ver. 3.4, Aspect Medical Systems). The EEG signal from the raw EEG port of the A-1050 monitor was introduced into a personal computer with Microsoft Windows ME. The BSA software (3) was used to analyze the data in real time. We used 2-s epochs, and each epoch overlapped the previous one by 75%. After applying a Blackman window function, the Fourier transform of each epoch was computed. We then calculated EEG bicoherence values from 3 min of EEG sample (360 epochs) using previously described equations (3).

The concentration of expired sevoflurane was continuously monitored using a Capnomac Ultima (Datex, Helsinki, Finland).

To avoid the noise of the electric cautery device during skin incision, all data were obtained at least 30 min after the start of surgery. Fentanyl IV was administered to minimize the influence of surgical stress on the EEG during surgery.

In the data collection period, the end-tidal concentration of sevoflurane was purposely maintained at set levels for 15 min and then changed to another concentration. Because the $t_{1/2 \text{ ke0}}$ for sevoflurane derived from the BIS data was 3.5 ± 2.0 , 15 min of maintenance time was considered to be enough to achieve a steady state. Because our previous study (4) revealed that bicoherence values showed 2 peaks in a fairly low-frequency (≤ 15.0 Hz) region along the diagonal plot ($f_1 = f_2$) of frequency versus frequency space, we defined $\text{aBIC}(f)$ as an average of bicoherence values (total, 11 points) in the area across the diagonal plot. Consequently, considering $\text{bic}(f_1, f_2) = \text{bic}(f_2, f_1)$, we calculated $\text{aBIC}(f)$ using the following equation. Here

Table 1. Demographic Data

Age (yr)	62 ± 9 (44–72)
Weight (kg)	56 ± 13 (34–80)
Height (cm)	160 ± 10 (140–172)
Male (n)	10
Female (n)	6

Values are mean $\pm$ SD (range).

$\text{bic}(f_1, f_2)$ is a raw bicoherence value calculated by our software, which is expressed as a percentage (0%–100%).

$$\begin{aligned} \text{aBIC}(f) = & 1/11[\text{bic}(f, f) + 2\{\text{bic}(f, f - 0.5) \\ & + \text{bic}(f + 0.5, f - 0.5) \\ & + \text{bic}(f + 0.5, f - 1.0) \\ & + \text{bic}(f + 1.0, f - 1.0) \\ & + \text{bic}(f + 1.0, f - 1.5)\}] \end{aligned}$$

In each analysis, $\text{aBIC}(f)$ values ($2.0 \leq f \leq 15.0$ Hz at each 0.5 Hz step) were calculated from 3 consecutive minutes of artifact-free EEG wave forms. Then we defined the maximum value among $\text{aBIC}(f)$ $2.0 \leq f \leq 6.0$ Hz as pBIC-low and the maximum value among $\text{aBIC}(f)$ $7.0 \leq f \leq 15.0$ as pBIC-high. We investigated the changes in both pBICs at 0.5%, 0.8%, 1.1%, 1.4%, 1.7%, 2.0%, and 2.3% sevoflurane in relation to BIS, SEF95, and burst suppression ratio. Bicoherence values for sevoflurane at 0.8%–2.3% were obtained during surgery; sevoflurane was initially maintained at 2.3% and then stepped down to 0.8%. When a “burst suppression” pattern was apparent on the EEG, we did not increase the sevoflurane concentration. If a BIS value during data collection was above 60, the sevoflurane concentration was increased to maintain the BIS value below 60 and further reduction of sevoflurane concentration was stopped to prevent the patient from waking. Lower concentration data were obtained during emergence from anesthesia.

MAP, HR, BIS, SEF 95, and pBICs values were analyzed by one-way analysis of variance followed by the Scheffé multiple comparison test. $P < 0.05$ was considered significant. Data are expressed as means $\pm$ SD.

RESULTS

Demographic data are shown in Table 1.

Table 2 shows the changes in MAP and HR during the study. MAP ($P = 0.07$) and HR ($P = 0.09$) did not change significantly. No patients needed phenylephrine to maintain MAP above 60 mm Hg during the data collection period. We observed a burst and suppression pattern in one patient at 2.3% sevoflurane. The data in this case was excluded from statistical analysis. In other cases, burst suppression ratios were below 5% during the study. All BIS values during the data collection period were below 60.

Figure 1 shows typical bicoherence patterns at each sevoflurane concentration in a 44-yr-old woman who underwent distal partial gastrectomy. At 0.5% sevoflurane, all bicoherence values were small. With increasing sevoflurane concentrations, bicoherence values showed

Table 2. Changes in MAP and HR

Sevoflurane concentration (%)	0.5	0.8	1.1	1.4	1.7	2.0	2.3
MAP (mm Hg)	91 ± 16	89 ± 14	83 ± 14	84 ± 13	81 ± 10	77 ± 13	81 ± 13
HR (bpm)	67 ± 19	64 ± 14	66 ± 13	65 ± 16	67 ± 16	69 ± 18	71 ± 14

Values are expressed as mean ± sd.

MAP = mean arterial blood pressure; HR = heart rate.

MAP and HR did not change significantly during the study.

2 peaks in a low-frequency region along the diagonal plot ($f_1 = f_2$) of frequency versus frequency space. The two peaks reached maximum at 1.1% or 1.4% sevoflurane. At a higher concentration of sevoflurane, the higher frequency peak became smaller; however, the lower frequency peak remained constant.

Figure 2 shows the changes in pBICs (pBIC-low and pBIC-high) at different sevoflurane concentrations. At 0.5% sevoflurane, both pBICs were low. As the sevoflurane concentration increased, pBIC-low increased significantly until it reached a plateau ($37.7\% \pm 7.5\%$) at 1.4% sevoflurane. Similarly, pBIC-high increased significantly when the sevoflurane concentration increased, reaching a peak ($35.1\% \pm 9.0\%$) at 1.4% sevoflurane.

At higher concentrations of sevoflurane, pBIC-high decreased.

Figure 3 shows the changes in BIS(A) and SEF 95(B) at different sevoflurane concentrations. BIS value decreased as the sevoflurane concentration increased until it reached a plateau at 1.4%. SEF 95, by contrast, decreased as the sevoflurane concentration increased up to 2.3%.

DISCUSSION

Our data show that both of the pBICs significantly changed with different concentrations of sevoflurane,

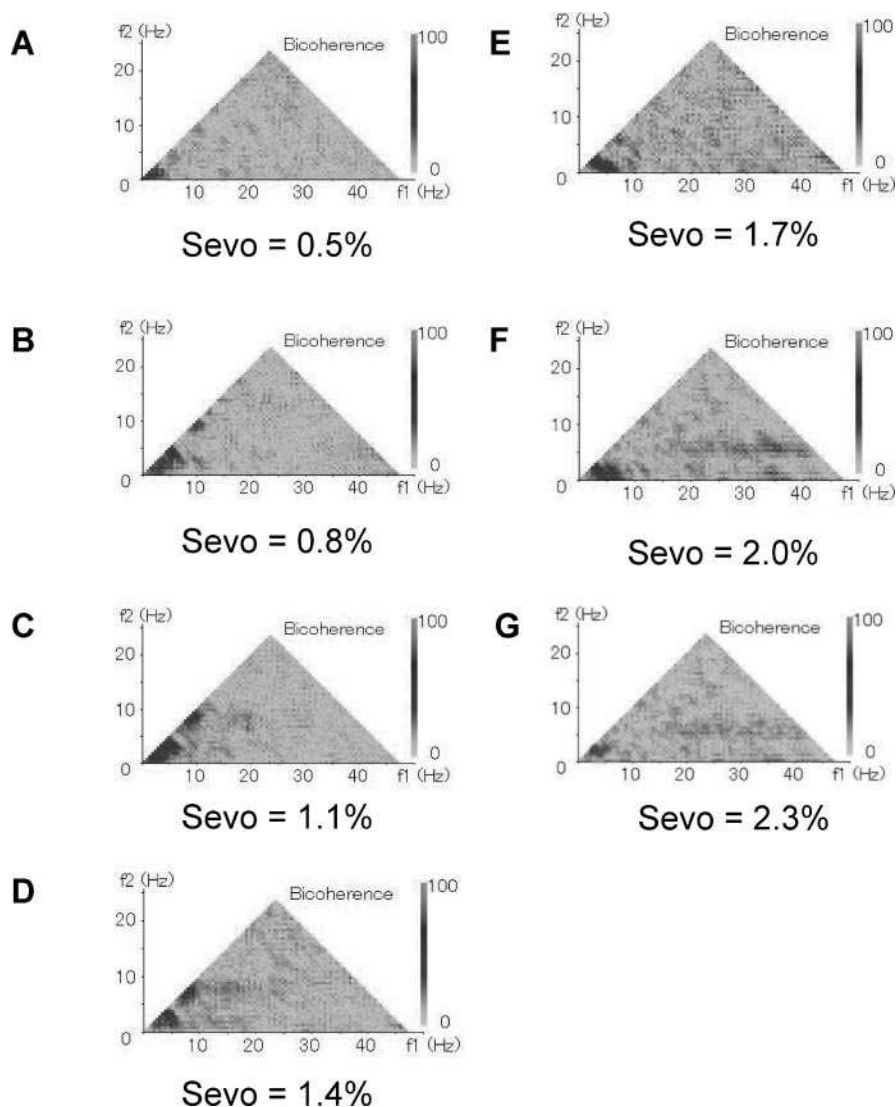


Figure 1. Typical patterns of electroencephalographic bicoherence in a 44-year-old woman who underwent distal partial gastrectomy. Bicoherence values (0%–100%) are plotted at end-tidal sevoflurane concentrations of 0.5%, 0.8%, 1.1%, 1.4%, 1.7%, and 2.0%.

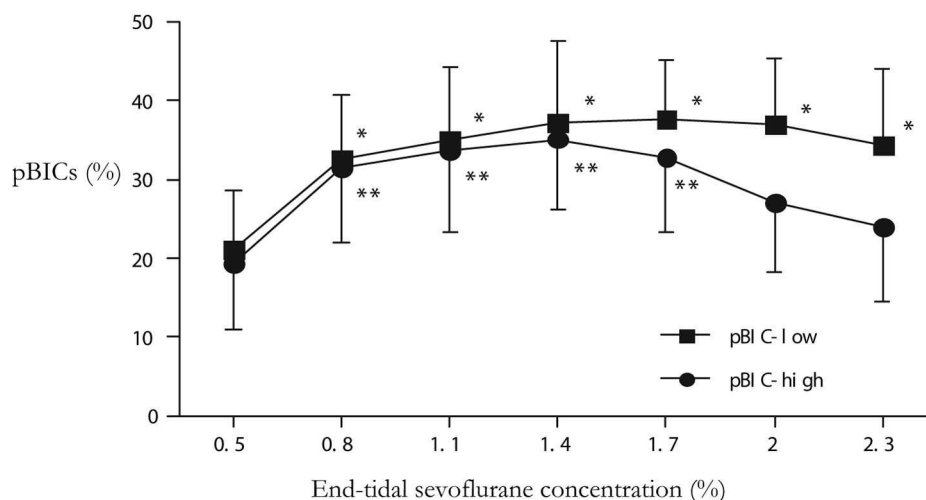


Figure 2. Relationship between the peak height of bicoherence (pBIC-low and pBIC-high) and sevoflurane concentration. *Significant difference in pBIC-low value compared with the value at 0.5% of sevoflurane. **Significant difference in pBIC-high value compared with the value at 0.5% of sevoflurane.

which suggests that the distribution pattern of bicoherence values is likely to be a good indicator for assessing the sedative effect of sevoflurane during surgery.

BIS decreased when the sevoflurane concentration increased up to 1.4%. However, BIS did not change between 1.4% and 2.3% sevoflurane. This limited reduction in BIS was also reported by Katoh et al. (5) during sevoflurane anesthesia and by Kurehara et al. (6) during isoflurane anesthesia. SEF decreased when the sevoflurane concentration increased up to 2.3%. Consistent with this result, we previously reported that SEF is superior to BIS for tracking the changes of anesthetic concentration during isoflurane-nitrous oxide anesthesia (7). BIS values between 40 and 65 have been recommended for general anesthesia (8). In this study, from 0.8%–2.3% of sevoflurane corresponded to the recommended BIS range. Because a dose-related change in BIS is limited, it may be difficult to titrate the sevoflurane concentration using BIS monitoring.

Similar to the BIS, both the pBIC-high and the pBIC-low plateaued at 1.4% sevoflurane. However, a decrease in pBIC-high followed a further increase in sevoflurane concentrations. For tracking changes of sevoflurane concentration, the combination of pBIC-high and pBIC-low was better than the BIS. In short, during light anesthesia, both pBICs are low. An increase in both pBICs indicates moderate or adequate anesthesia. High pBIC-low and low pBIC-high might indicate too deep anesthesia.

In our previous study, the changes in the two EEG bicoherence peaks were also investigated during isoflurane and epidural anesthesia (4). When the isoflurane concentration increased, pBIC-low increased until it reached a plateau at 0.9% isoflurane. Similarly, pBIC-high increased significantly as the isoflurane concentration increased, reaching a maximum at 0.9% isoflurane and after this, at a higher isoflurane concentration ($\geq 1.3\%$), pBIC-high decreased slightly. The changes in pBICs during isoflurane anesthesia were similar to those observed during sevoflurane anesthesia. Schwender et al. (9) observed a dose-dependent

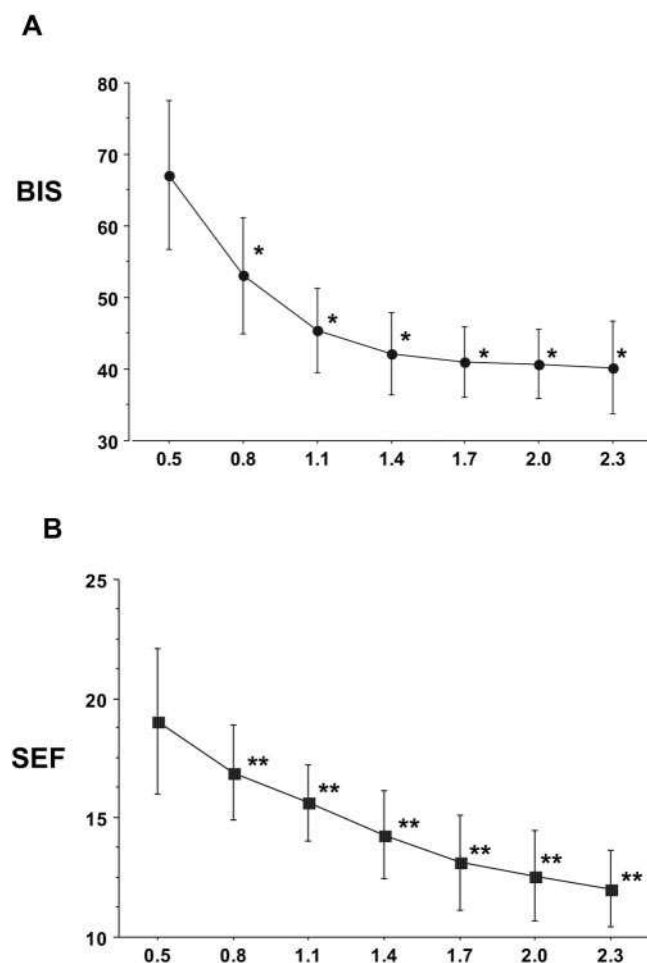


Figure 3. The changes in Bispectral Index (BIS) (A) and spectral edge frequency (SEF) 95(B) at different sevoflurane concentrations. *Significant difference in BIS value compared with the value at 0.5% of sevoflurane. **Significant difference in SEF 95 value compared with the value at 0.5% of sevoflurane.

decrease in SEF with increasing end-expiratory concentrations of isoflurane, desflurane, and sevoflurane. The minimum alveolar concentration equivalent dose of these volatile anesthetics was associated with equipotent EEG and central nervous system suppression. These results suggest that concentration-dependent changes in EEG are similar in both sevoflurane and isoflurane anesthesia.

PBIC-high was observed at 7–14 Hz of frequency, which is the frequency of spindle waves, and pBIC-low was observed at around 4 Hz, the frequency of δ waves (4). Therefore, we have surmised that the changes in pBIC-high indicate spindle wave activity and the changes in pBIC-low indicate δ wave activity. EEG spindle wave pattern typically occurs during sleep and is also predominant during light anesthesia; however, in a deeper anesthesia, spindle waves become smaller and the δ waves become predominant. Steriade et al. (10) investigated the pacemakers of these two EEG waves and concluded that the pacemaker of the spindle wave was thalamo-cortico-thalamic circuits and the rhythm of the δ wave was the intrinsic rhythm of thalamocortical neurons. In essence, as anesthetic concentrations increase, spontaneous cerebral cortex activity decreases, and the intrinsic synchronized rhythm of thalamocortical neurons predominates the EEG. Increase in pBIC-low and pBIC-high might indicate this increase in EEG synchronization driven by the thalamus.

In the current study, we used fentanyl to prevent pain from surgical manipulation. Therefore, the effect of fentanyl on the EEG should be considered. Fentanyl causes a progressive slowing of the EEG. The serum concentration that caused half of the maximal EEG slowing was reported as 6.9 ± 1.5 ng/mL (11). During the data collection period, the effect-site concentration calculated by Tivatrainer (<http://www.eurosiva.org/>) was approximately 2 ng/mL, which is far lower than the concentration that affects EEG. In our previous study, 3 μ g/kg of fentanyl did not affect pBICs, BIS, and SEF95 (12). The effect of fentanyl on EEG might be neglected in the present study because the doses of fentanyl (2 μ g/kg bolus plus continuous infusion of 2 μ g $\cdot$ kg⁻¹ $\cdot$ h⁻¹) used were too small to have an effect on EEG. However, the level of surgical stimulation was not constant during the data collection period, and

surgical stimulation can also influence EEG. Because these effects on EEG in clinical settings cannot be completely excluded, further volunteer studies would be needed to confirm the usefulness of bicoherence.

In summary, both pBIC-low and pBIC-high significantly increased with increasing sevoflurane concentrations up to 2.3% (end-tidal). BIS decreased as the sevoflurane concentration increased up to 1.4%. PBIC-low also plateaued at 1.4% sevoflurane. A decrease in pBIC-high reflected a further increase in sevoflurane concentration. We conclude that combined use of pBIC-high and pBIC-low is a valid and better indicator than BIS for assessing anesthetic state during sevoflurane anesthesia.

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